Azmal Fuad Hamzah

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Academic Profile

A highly motivated undergraduate student pursuing a B.Sc. (Hons.) in Mathematics at Kishoreganj University, with a strong foundation in Computational Fluid Dynamics (CFD), numerical methods, and mathematical modeling. Experienced in using simulation tools like COMSOL, ANSYS, and MATLAB to solve complex heat transfer and fluid flow problems. Aspiring to pursue an MSc in Applied Mathematics or Data Science, with a goal to build a career in advanced research, academia, or industrial R&D.

Education

Kishoreganj University February 2022 – May 2027 (Expected)
B.Sc. (Hons.) in Mathematics, CGPA: 3.63/4.00 Kishoreganj, Bangladesh

- **1st Year:** GPA 3.33 (1st Sem: 3.15, 2nd Sem: 3.50)
- **2nd Year:** GPA 3.73 (1st Sem: 3.67, 2nd Sem: 3.80)
- **3rd Year:** GPA 3.82 (1st Sem: 3.79, 2nd Sem: 3.85)

Sristy College of Tangail 2021
Higher Secondary Certificate (HSC), Group: Science, GPA: 4.83/5.00 Dhaka Board

Kalta Avoya Charan High School 2019
Secondary School Certificate (SSC), Group: Science, GPA: 4.06/5.00 Dhaka Board

Research Interests

Computational Fluid Dynamics (CFD), Heat Transfer, Hydrodynamics, Fluid Flow and Vortex Shedding, Numerical Simulations, and Energy Cost Optimization.

Research Experience

Undergraduate Thesis / Course Project Ongoing
Supervisor: Md. Abadur Rahman, Lecturer, Department of Mathematics, Kishoreganj University

- **Topic:** CFD analysis of fluid flow and heat transfer around confined geometries (semi-circular cylinders and airfoil-shaped channels).
- Investigating the onset of vortex shedding, Reynolds and Prandtl number effects, and heat transfer characteristics of hybrid nanoparticles in Newtonian fluids.
- Utilizing **COMSOL Multiphysics, ANSYS, and MATLAB** for numerical modeling, meshing, and simulation.

Independent Research Project Ongoing

- **Topic:** Thermal Comfort Optimization and Energy Consumption Analysis in Enclosed Spaces.
- Developing a numerical model to optimize heater placement, cycle time, and electrical cost while maintaining a fixed temperature boundary.
- Investigating heat loss through structural openings (doors/windows) and simulating temperature decay dynamics to minimize energy consumption.

Technical Skills

Simulation & Modeling:	COMSOL Multiphysics, ANSYS, MATLAB
Programming Languages:	C, Python, C++ (basic), Fortran (basic)
Other Tools:	GitHub, LaTeX, Microsoft Office, EndNote, Zotero
Soft Skills:	Leadership, Event Management, Public Speaking, Time Management

Certifications

CS50x: Introduction to Computer Science Harvard University (via edX) Expected: [Month, 2026]

Relevant Coursework

Numerical Methods, Mathematical Modeling, Differential Equations (ODE & PDE), Fluid Dynamics, Real Analysis, Linear Algebra, Complex Analysis, Probability & Statistics, Programming (C/C++), MATLAB.

Leadership & Extracurricular Activities

Debating Club, Kishoreganj University
Treasurer 2022 – Present

Pi Club, Kishoreganj University
Organizing Secretary 2024 – 2025

Rover Scout, Kishoreganj University
Group Leader 2022 – Present

Awards & Achievements

National Undergraduate Math Olympiad Participant

Work Experience

Byangachi Magazine
Proofreader October 2020 – December 2020
Involved in proofreading and correcting manuscripts prior to publication.

Languages

Bangla (Native), English (Fluent)

References

Md. Abadur Rahman
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Md. Saiful Islam
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